SMGT 430/530 Introduction to Sports Analytics
The Big Dance

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1 Introduction

With the NCAA women's basketball playoff tournament just around the corner, we investigated the 3-point shooting talent of all players in the NCAA to determine the best shooter in all of college basketball adjusted for quality of opposition and conference. An accurate assessment of shooting ability is for example essential for scouts in the WNBA trying to fill up roster spots and deciding which player to pick in the draft. While raw three point shooting percentage is a good starting point, it can be easily influenced by luck, sample size and strength of schedule. In this report, we distinguish between true elite shooters and those who benefited from favorable match-ups, easy schedules, or simply luck to give front offices a better understanding of the underlying true player talent.

2 Methods

This analysis models player's three point outcomes using a hierarchical or mixed effects binomial logistic regression model instead of relying on observable three point percentage. We estimate the true probability of a successful three-point field goal. The model is specified as follows:

$$y_{ijkl} \sim \text{Binomial}(n_{ijkl}, p_{ijkl}) \quad \text{and} \quad \text{logit}(p_{ijkl}) = \mu + \theta_i + \gamma_j + \delta_k + \phi_l$$

where y_{ijkl} is the number of made three-pointers out of n_{ijkl} attempts, and p_{ijkl} is the underlying probability of making the shot. The parameter μ represents the NCAA baseline average. The term θ_i is the individual player effect, while γ_j is the player's conference effect. To incorporate strength of schedule, we included δ_k for the opposing team effect and ϕ_l for the opposing conference effect. Finally, to calculate each player's true talent, we neutralized the other effects by only summing the global baseline and the player effect $(\mu + \theta_i)$.

3 Results

Our analysis reveals the following top 10 three-point shooters based on pure talent (Table 1), with Allie Turner leading the NCAA according to our model. Figure 2 shows the observed three-point percentage on the x-axis versus the estimated true talent on the y-axis. Overall, these results seem reasonable. The model recognizes that high-volume shooters see very little shrinkage, proving the model correctly weighs sample size and

environmental context such as playing in an easier conference. We see that Emilee Fox boasts the highest observed shooting-pct, but only ranks as 9th since she only took 104 shots in addition to playing in a mid-major conference.

Top 10 Three-Point Shooters

NCAA Women's Basketball 2025-26 Season

Rank	Player	Team	Games	3PA	Observed %	True Talent %
1	Allie Turner	Gonzaga	31	171	45.6	38.3
2	Katie Donovan	Davidson	30	139	46.8	38.3
3	Azzi Fudd	UConn	31	213	45.1	38.3
4	Ellie Brueggemann	Lindenwood	30	193	44.0	37.7
5	Elle Evans	Kansas	30	139	47.5	37.7
6	Meghan Andersen	Fairfield	30	202	42.1	37.6
7	Avery Mills	Liberty	22	157	44.6	37.6
8	Natalie Pasco	Boise St	30	167	44.3	37.5
9	Emilee Fox	S Dakota St	24	104	49.0	37.3
10	Cali Smallwood	UAB	28	206	41.7	37.1

Figure 1: NCAA top 10 three point shooters based on the true talent estimated by our hierarchical model

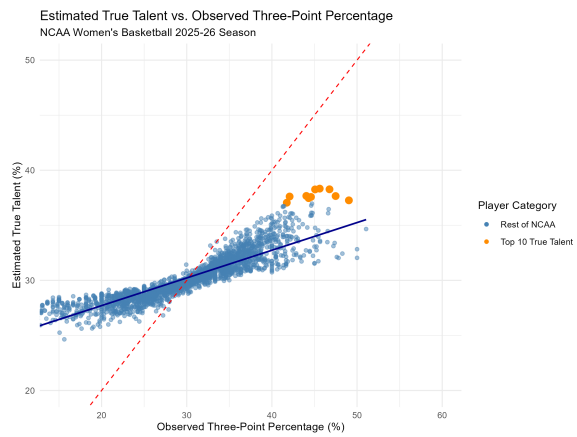

Figure 2: Estimates true talent vs. observed three-point percentage

4 Discussion

I believe that Azzi Fudd should be the highest ranked shooter on the list, since she shoots an absurd 45.1% from three while being on arguably the best team in the country UConn. She also does this on the most attempts (213), which is why I would put her true talent percentage as 42% thus ranking as the highest on the list. Since hierarchical models follow an underlying normal distribution, outliers like her get pulled in even more. Additionally, as one of the star players, she is likely taking more contested and off the dribble attempts, making her 45.1% even more impressive.

Ultimately, the purpose of this report is to evaluate true talent three point percentage in NCAA women's basketball, which is an important metric for draft and scouting analysis and decision making during drafting and free-agency. Using a hierarchical mixed-effects model, we strip away the noise of varying sample sizes and environmental influences, solely focusing on shooting talent. We identified Allie Turner, Katie Donovan, and Azzi Fudd to be the best shooters of the season. Unlike professional basketball, college basketball features massive differences in defensive quality. Relying on raw percentages might be misleading, as a mid-major standout might simply be benefiting from weak defenses, while a true sniper is hidden by a tough conference. By contextualizing these environmental variances, front offices can confidently draft shooters whose talent will actually translate to the professional level.